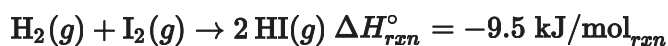
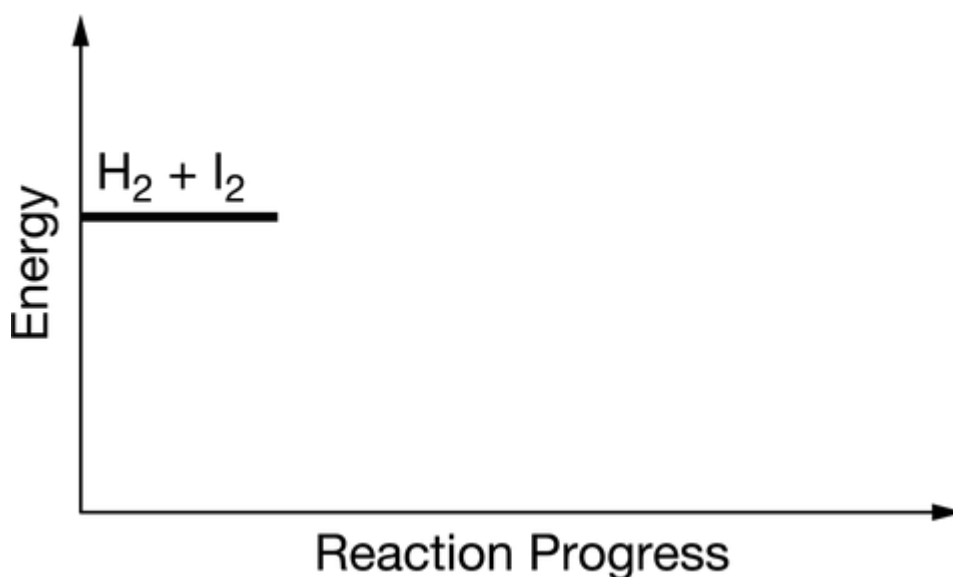


Unit 5 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



$\text{H}_2(g)$ and $\text{I}_2(g)$ can react to produce $\text{HI}(g)$, as represented above. Shown below is an incomplete energy diagram for the reaction. The potential energy of the reactants is indicated on the diagram.



- (a) Carefully complete the energy diagram by drawing a curve that accurately shows the progress of the reaction, beginning at the reactants, moving through the transition state, and ending at the final products. When adding the products to the diagram, be sure to use an energy level that is appropriate relative to the energy level of the reactants.



Please respond on separate paper, following directions from your teacher.

- (b) On your energy diagram, draw a dashed curve to show the reaction progress from reactants to products in the presence of a suitable catalyst for the reaction.



Please respond on separate paper, following directions from your teacher.

- (c) On your energy diagram, draw a vertical line segment with a length that corresponds to the



Unit 5 Progress Check: FRQ

activation energy for the reverse uncatalyzed reaction: $2 \text{HI}(g) \rightarrow \text{H}_2(g) + \text{I}_2(g)$.



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

		✓
0	1	2

The drawing meets both of the criteria below:

- ☐ There is a horizontal line drawn on the right side of the graph that is lower than the horizontal line that represents the reactants.
- ☐ There is a curve drawn between the two horizontal lines that starts at the right end of the reactant line, reaches a peak that is higher than both horizontal lines, and ends at the left end of the product line.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

	✓
0	1

The response shows a dashed curve that starts and ends at the same points as the curve drawn for part (a) but has a shorter peak.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. To earn credit, the response must be consistent with the graph drawn in part (a).



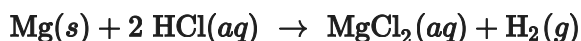
Unit 5 Progress Check: FRQ



0	1
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The response shows a vertical line segment with its bottom end at the level of the product line and its top end at the level of the top of the peak for the uncatalyzed reaction drawn on the graph.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



In an experiment, a student places a small piece of pure $\text{Mg}(s)$ into a beaker containing **250. mL** of **6.44 M HCl(aq)**. A reaction occurs, as represented by the equation above.

- (a) Write the balanced net ionic equation for the reaction between $\text{Mg}(s)$ and $\text{HCl}(aq)$.

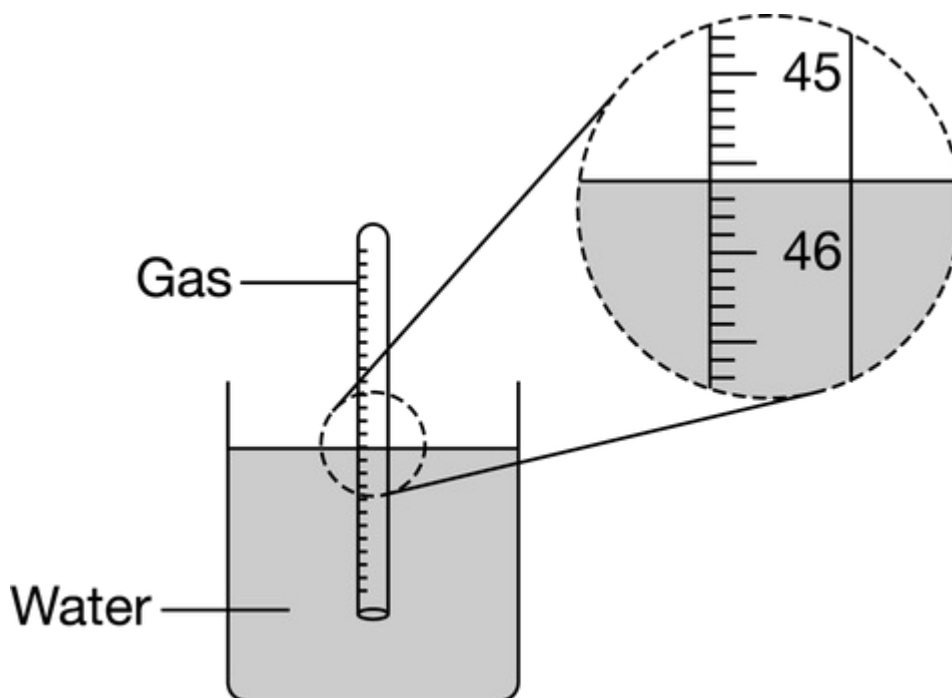


Please respond on separate paper, following directions from your teacher.

- (b) The student collects the $\text{H}_2(g)$ produced by the reaction and measures its volume over water at **298 K** after carefully equalizing the water levels inside and outside the gas-collection tube, as shown in the diagram below. The volume is measured to be **45.6 mL**. The atmospheric pressure in the lab is measured as **765 torr**, and the equilibrium vapor pressure of water at **298 K** is **24 torr**.



Unit 5 Progress Check: FRQ



Calculate the following.

- (i) The pressure inside the tube due to the $\text{H}_2(g)$



Please respond on separate paper, following directions from your teacher.

- (ii) The number of moles of $\text{H}_2(g)$ produced in the reaction



Please respond on separate paper, following directions from your teacher.

- (c) In a different experiment, a student places a piece of pure $\text{Ba}(s)$ in a beaker containing $250. \text{ mL}$ of $6.44 \text{ M HCl}(aq)$ and observes that the rate of appearance of $\text{H}_2(g)$ bubbles when $\text{Ba}(s)$ was added is much faster than the rate of appearance of bubbles when the $\text{Mg}(s)$ was added. The student claims that the greater reactivity is due to the difference in ionization energy between Mg and Ba . Explain the difference in ionization energy between Mg and Ba in terms of atomic structure.



Please respond on separate paper, following directions from your teacher.

HCl can be produced by the exothermic reaction $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$. A proposed



Unit 5 Progress Check: FRQ

mechanism for the reaction has two elementary steps as shown below.

Step 1:	$\text{H}_2(g) + \text{ICl}(g) \rightarrow \text{HI}(g) + \text{HCl}(g)$	(slow)
Step 2:	$\text{HI}(g) + \text{ICl}(g) \rightarrow \text{HCl}(g) + \text{I}_2(g)$	(fast)

(d) Write a rate law for the overall reaction that is consistent with the proposed mechanism.

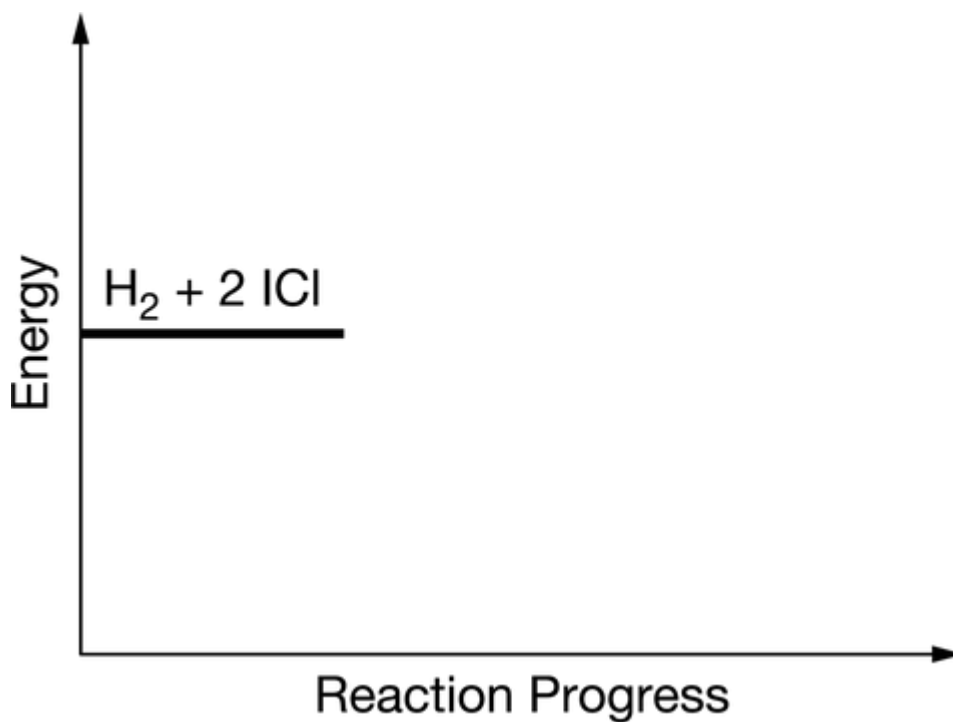


Please respond on separate paper, following directions from your teacher.

(e) On the incomplete reaction energy diagram below, draw a curve that shows the following two details.

The relative activation energies of the two elementary steps

The enthalpy change of the overall reaction



Please respond on separate paper, following directions from your teacher.

Part (a)

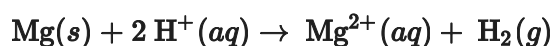
Unit 5 Progress Check: FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives the following equation (phases not required):

**Part (b)(i)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives the following calculation:

$$765 \text{ torr} - 24 \text{ torr} = 741 \text{ torr}$$

Part (b)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The points in part (b)(ii) can be earned if an incorrect pressure from part (b)(i) is used correctly in part (b)(ii).



0	1	2
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The response gives one of the following calculations with no errors:

☐ $PV = nRT$

$$n = \frac{PV}{RT} = \frac{\left(\frac{741}{760} \text{ atm}\right)(0.0456 \text{ L})}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 1.82 \times 10^{-3} \text{ mol (to three significant figures)}$$



Unit 5 Progress Check: FRQ

☐ $PV = nRT$

$$n = \frac{PV}{RT} = \frac{(741 \text{ torr})(0.0456 \text{ L})}{(62.4 \text{ L torr mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 1.82 \times 10^{-3} \text{ mol (to three significant figures)}$$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
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The response meets both of the following criteria:

- ☐ The response indicates that **Ba** has a lower ionization energy than **Mg** does.
- ☐ The response indicates that the valence electrons in **Ba** are in a shell that is farther from the nucleus, so the valence electrons in **Ba** have less attraction for the nucleus than do the valence electrons in **Mg** and are thus easier to remove.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1
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The response gives the following rate law:

☐ $\text{Rate} = k[\text{H}_2][\text{ICl}]$

Part (e)



Unit 5 Progress Check: FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3
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The drawn plot meets all three of the following criteria:

- ☐ There is a curve that starts at the right end of the reactant line, goes up to form one or more peaks, and then goes down and ends at the level of the energy of the product.
- ☐ The curve has two peaks, the first higher than the second.
- ☐ The drawn plot has a horizontal line representing the energy of the product that is lower than the line for the reactants.